

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Eighth Semester B.E. (Credit System) Degree Examinations
 May - June 2023

19BT8X40 - BIOFUEL ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Draw diagrams wherever necessary.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Elaborate on the role of biofuels in mitigating detrimental effects of fossil fuels.	5	L*1	1	1
	b) Explain how energy is produced by utilizing wood residues.	5	L2	1	1
	c) Describe the process of biodiesel production with equation.	10	L2	2	1,2
2.	a) Write a brief account on any two starch crops for biofuel production.	5	L1	1	1
	b) Explain sewage as a source for biofuel production.	5	L2	1	1
	c) Explain any five quality control tests adopted for Biodiesel fuel property standardization.	10	L2	2	1,2
3.	a) Elaborate on forest crops as resource available for bioenergy production.	10	L2	1	1
	b) Describe the fermentation process of bioethanol production with flow chart.	10	L2	2	2
Unit – II					
4.	a) Explain of direct and indirect bio-photolysis method of hydrogen production.	10	L3	3	2
	b) Explain the basic principle on which Microbial Fuel Cell (MFC) works with biochemical equations.	10	L2	4	2
5.	a) Describe the factors that play a major role in hydrogen production.	10	L2	3	1
	b) Briefly explain the importance of Anode and Cathode compartment of MFC.	10	L2	4	2
6.	a) Explain any two integrated continuously stirred tank reactor configurations for bio-hydrogen production.	10	L2	3	2
	b) Explain with suitable equations power production concept for MFC.	10	L3	4	1
Unit – III					
7.	a) Explain anaerobic digestion and elaborate on two main types of digesters.	10	L2	5	2,6
	b) Elaborate on incineration facility and technology planning.	10	L2	5	2
8.	a) Write briefly about mass burning and refuse derived fuel (RDF) waste disposal systems.	10	L2	5	2
	b) Explain pyrolysis and gasification processes.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
